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Research article

Lightweight network study of leather defect segmentation with Kronecker product multipath decoding

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Abstract: In the leather production process, defects on the leather surface are a key factor in the quality of the finished leather. Leather defect detection is an important step in the leather production process, especially for wet blue leather. To improve the efficiency and accuracy of detection, we propose a leather segmentation network using the Kronecker product for multi-path decoding and named KMDNet. The network uses Kronecker products to construct a new semantic information extraction layer named KPCL layer. The KPCL layer is added to the decoding network to form new decoding paths, and these different decoding paths are combined that segment the defective part of the leather image. We collaborate with leather companies to collect relevant leather defect images; use Tensorflow for training, validation, and testing experiments; and compare the detection results with non-machine learning algorithms and semantic segmentation algorithms. The experimental results show that KMDNet has a 1.99% improvement in $F1$ score compared to UNet for leather and a nearly three times improvement in detection speed.

Keywords: wet blue leather defect; semantic segmentation; multi-path decoding; Kronecker product; feature fusion

1. Introduction

Leather is one of the most essential raw materials for household items in people's lives today, for example, fur clothing, leather shoes, leather seats, etc. Most studies on leather defect detection algorithms are based on computer vision and mainly focus on finished leather [1–7], but there are fewer studies on wet blue leather. Compared with finished leather, wet blue leather is still in a semi-processed state, its leather is moister and its surface is relatively uneven. So the captured images of